ARMAN POUYAEI, PhD

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Professional Summary

Senior data engineer and data scientist with 10+ years building production machine-learning and predictive-modeling systems on large, high-dimensional data. PhD-trained in statistical inference, Bayesian estimation, and optimization. Currently leads a global severe-storm hazard modeling program end to end — AI-based weather emulation, GPU-accelerated ensemble forecasting, extreme value theory, physics-based transport and emission modeling, and forward-looking scenario design — from solution architecture through production delivery. Core discipline: extracting predictive signals from large weather datasets and validating them by hindcast (backtest) evaluation on forecasts as issued, not after-the-fact analyses. Works fluently in Python across HPC and cloud at petabyte scale, leads and mentors engineering teams, and translates technical findings into plain-language recommendations through visualization, prototypes, and executive briefings. 30+ peer-reviewed publications.

Technical Skills

Statistical & Predictive Modeling: Regression, classification, clustering, Bayesian inference and state estimation, time-series forecasting, extreme value theory (GEV/GPD, peaks-over-threshold) and return-period estimation, probabilistic and ensemble forecasting, uncertainty quantification, forecast verification and skill scoring, hindcast (backtest) validation, hypothesis testing, constrained optimization.

Machine Learning & AI: PyTorch, JAX, scikit-learn, TensorFlow; XGBoost, Random Forest, CNN/UNet, GNN, PINN, neural operators and diffusion-based generative models; AI weather emulators; multi-GPU distributed training and inference; LLM and NLP tooling, agentic and retrieval workflows.

Physical & Numerical Modeling: Numerical weather prediction output and large weather datasets (satellite, radar and station observations; ERA5 and MERRA-2 reanalyses), Lagrangian particle transport and dispersion models, physics-based emission schemes, data assimilation (3DVAR, EnKF), climate and atmospheric-chemistry models, downscaling, bias correction, scenario and climate-pathway analysis, climate teleconnection indices (ENSO, MJO, NAO).

Programming & Data: Python (NumPy, pandas, Xarray, Dask, GeoPandas), R, SQL, Bash, Fortran; GRIB2, BUFR, NetCDF, HDF5, Zarr, Parquet; relational and columnar stores; schema, warehouse, and data-model design.

Engineering & Scale: AWS (Batch, S3, Lambda), Docker, SLURM/MPI, CUDA and multi-GPU clusters; production ETL, automated QA/QC, anomaly detection, reproducible pipelines.

Visualization & Communication: Tableau, Matplotlib, Plotly, Seaborn, interactive HTML dashboards, PowerPoint; prototypes and demonstrations built for non-technical stakeholders and executive audiences.

Professional Experience

Senior Data Engineer (Data Science) Aug 2026 – Present

MSCI Inc., New York, NY

Severe-storm hazard modeling, predictive analytics, and AI platform engineering for climate risk. (First Street acquired by MSCI, August 2026.)

- Lead the severe-storm modeling program end to end — blizzard, dust storm, and sandstorm hazards — owning solution architecture, model and tool development, GPU-accelerated training and inference, validation, and production delivery.
- Built a global ensemble forecasting capability on AI-based weather emulators, generating large perturbed-initial-condition ensembles on GPUs at a fraction of the cost of physics-based numerical weather prediction, then post-processing members into calibrated hazard probabilities and spread-based confidence measures; validated by hindcast (backtest) evaluation against independent observations — scoring forecast error, ensemble spread, and calibration — so reported skill reflects what the system knew at issue time, not after-the-fact analyses.
- Applied extreme value theory (GEV and generalized Pareto, peaks-over-threshold) to fit the tails of storm severity distributions and derive return-period estimates well beyond the observational record, with bootstrapped confidence intervals attached to every return level.
- Designed a forward-looking scenario framework that projects hazard frequency and severity onto future climate pathways, so present-day risk and mid- to late-century outcomes can be compared on a consistent, auditable basis.
- Developed a Lagrangian particle transport model coupled to a physics-based dust and sand emission scheme — friction-velocity thresholds with soil, vegetation, and land-surface controls — to simulate emission, long-range transport, and deposition at the spatial resolution needed for asset-level exposure.
- Build, tune, and validate machine-learning models (PyTorch, JAX, scikit-learn) that extract predictive signals and score risk exposure across hundreds of millions of records — owning feature engineering, training, cross-validation, and production deployment end to end.
- Set technical direction across the program and review engineering and analytical work; keep the team aligned to milestones through regular check-ins with engineers and analysts, and report delivery status to senior leadership.
- Develop AI and LLM-based tooling and agentic workflows that automate model diagnostics, output review, and documentation; partner with product, research, and client-facing teams to turn open-ended business questions into measurable requirements and tangible prototypes.

Data Engineer / Research Scientist May 2025 – Aug 2026

First Street, New York, NY

Machine learning and data pipeline engineering for global climate risk analytics.

- Designed and deployed end-to-end ML workflows covering data preparation, training, validation, and iterative QA/QC for global predictive models, including tuning and scaling UNet-based emulators.
- Built an automated quality-control and anomaly-detection framework that evaluates model outputs against independent benchmarks and surfaces failures through dashboards — cutting review time during global model release cycles.
- Architected containerized ETL on AWS Batch serving petabyte-scale datasets (Zarr, Parquet, NetCDF) to downstream analytics and client products.

- Converted exploratory notebook research into production-grade, reproducible pipelines on SLURM clusters and AWS, working alongside the science team to preserve analytical intent.

Postdoctoral Research Associate Mar 2023 – May 2025
NOAA GFDL / Princeton University
Cooperative Institute for Modeling the Earth System (CIMES), Princeton University.
- Applied statistical and machine-learning analysis to decadal, high-dimensional datasets to isolate driver–response relationships hidden in noisy signals; findings published in *Geophysical Research Letters*.
- Led a modeling workstream spanning four research groups and 8+ scientists — defining scope, sequencing deliverables, and keeping distributed contributors aligned to a shared roadmap.
- Built and benchmarked predictive model components against independent observational datasets, documented uncertainty for downstream users, mentored graduate students, and delivered invited talks at NASA GISS and AGU.

Postdoctoral Research Fellow Sep 2022 – Feb 2023
University of Houston
Bayesian inference, inverse modeling, and predictive analytics.
- Designed a 3DVAR variational inference framework that assimilates high-volume observational streams into predictive models — a large-scale optimization and state-estimation problem over sparse, heterogeneous data.
- Constrained model uncertainty by fusing satellite and in-situ measurement sources; authored competitive research proposals to DOE, NOAA, NASA, and NSF and supervised undergraduate and graduate researchers.

Graduate Research Assistant 2018 – 2022
University of Houston
Air Quality Forecasting and Modeling Lab — statistical modeling, data fusion, and satellite data analysis.
- Built and open-sourced a Lagrangian particle-trajectory framework (C-TRAIL v1.0) that traces how quantities move through large spatiotemporal simulations and attributes downstream outcomes to specific upstream sources; adopted by external research groups and extended into a diagnostic tool that ranks each source region's contribution during extreme episodes.
- Developed data-fusion operators that reconcile heterogeneous observational sources — satellite, ground-network, and aircraft measurements with differing resolution, bias, and coverage — within a Bayesian assimilation framework, producing constrained estimates from sparse and inconsistent inputs.
- Designed a physics-based scheme for processes below the model's resolved grid scale, measurably improving predictive skill (JAMES 2021), and co-developed deep-learning surrogates, spatial-imputation models, and super-resolution bias correction with the group's machine-learning team.

Graduate Research Assistant 2014 – 2017
University of Tehran
- Designed optimization and predictive-modeling frameworks for renewable energy systems; published in *Energy Conversion and Management* and IEEE.

Education

Ph.D., Atmospheric Science 2018 – 2022
University of Houston, TX
Quantitative focus: Bayesian data assimilation, numerical modeling, statistical analysis of large observational datasets.
M.Sc., Renewable Energies Engineering 2013 – 2016
University of Tehran, Iran
B.Sc., Mechanical Engineering 2008 – 2012
University of Shiraz, Iran
Awards: Outstanding Graduate Work Scholarship, University of Houston (2021 & 2022); full scholarships for M.Sc. and B.Sc.

Leadership, Mentoring & External Engagement

- **Team & Project Leadership:** Coordinate cross-functional delivery at MSCI and First Street through regular working sessions where engineers and analysts walk through progress and surface blockers together. At Princeton/NOAA GFDL, directed a modeling workstream across four research groups and 8+ scientists, with collaborators at NASA, NCAR, Brookhaven, and PNNL.
- **Mentoring, Teaching & Funding:** Mentored undergraduate and graduate researchers at Princeton/NOAA GFDL and the University of Houston, plus two international high school students through the Collegiate Mentorship Program (2024). Teaching assistant for Dynamic Meteorology and Principles of Atmospheric Science. Authored proposals to DOE, NOAA, NASA, and NSF.
- **Communicating to Mixed Audiences:** Invited speaker, NASA GISS Seminar Series (2025) and NOAA GFDL Seminar Series (2024). Conference presenter at AGU Fall Meetings (2021–2024), AMS Annual Meeting (2024), and CMAS (2021).
- **Professional Service & Outreach:** Lead convener, AGU 2026 session on aerosol impacts on climate, air quality, and health; convener of three AGU sessions across 2024 and 2025. Peer reviewer for JGR-Atmospheres, *Atmospheric Environment*, *Scientific Reports*, and six additional journals. Organizing committee, Spring Into Science science outreach, Princeton University (2024).

Selected First-Author Publications (7 first-author of 30+ total)

- Pouyaei, A. et al. (2026, under review). Linking Prognostic Fire Emissions to Atmospheric Chemistry and Aerosols in GFDL's Atmospheric and Land Model version 4.2 (AM4.2/LM4.2). *J. Advances in Modeling Earth Systems.* https://doi.org/10.22541/essoar.15005375/v1
- Pouyaei, A. et al. (2025). ENSO-driven Variability in Ozone Sources and Its Impact on Tropospheric Ozone Radiative Forcing. *Geophysical Research Letters*, 52(15).
- Pouyaei, A. et al. (2025). Implementation of Dynamic Fire Injection Height in GFDL's AM4.0: Impacts on Aerosol Profiles and Radiation. *J. Advances in Modeling Earth Systems*, 17(4).
- Pouyaei, A. et al. (2023). Downwind Ozone Changes of the 2019 Williams Flats Wildfire: WRF-Chem/DART Assimilation of OMI NO_2, HCHO, and MODIS AOD. *J. Geophysical Research: Atmospheres*, 128(9).
- Pouyaei, A. et al. (2020). C-TRAIL Model v1.0 derived from CMAQ Model v5.2. *Geoscientific Model Development*, 13(8).